Infinity Learn

OLYMPIAD WORKSHEET

Heat

Science — Physics · Class 7 · Max Marks: 17 · Time: 45 min

Section A — Multiple Choice (Higher Order) (1 × 14 = 14)

1. Two identical cups of hot water are taken; one is wrapped in a woollen cloth. After 10 minutes, the water that is hotter is in the:

A. neither — both are equal
B. cup wrapped in wool
C. bare cup
D. cup that was stirred

2. On a cold morning an iron bench and a wooden bench are at the same temperature, yet the iron feels colder. This is because iron:

A. conducts heat away from the hand faster
B. is heavier
C. radiates cold
D. is actually colder

3. Heated over the same flame for the same time, water gets hottest in a:

A. shiny silver metal pot
B. glass beaker
C. shiny white ceramic pot
D. dull black metal pot

4. The temperature of a body is best described as a measure of:

A. its mass
B. its volume
C. the average degree of hotness of its particles
D. the total heat energy stored in it

5. The shiny silvered walls of a thermos flask mainly reduce heat loss by:

A. radiation
B. conduction
C. convection
D. all three equally

6. Rods of copper, aluminium, iron and glass of equal size are heated at one end with a wax ball at the other. The ball that falls LAST is on the:

A. iron rod
B. glass rod
C. copper rod
D. aluminium rod

7. During the day a sea breeze sets in because the air over the land:

A. becomes cooler
B. gets warmer and rises
C. sinks down
D. becomes denser

8. Loose white cotton clothes are comfortable in summer because white reflects heat and loose cotton also:

A. soaks up sweat and lets air flow (convection)
B. radiates heat inward
C. blocks all air
D. conducts heat to the body

9. Which arrangement keeps a block of ice from melting the longest?

A. ice in a metal box
B. ice wrapped in a woollen blanket
C. ice on an open steel plate
D. ice on a black tray in sunlight

10. The only mode of heat transfer that can take place through a vacuum is:

A. conduction and convection
B. convection
C. radiation
D. conduction

11. **A laboratory thermometer generally has a range of about:**

A. 0°C to 50°C
B. 100°C to 250°C
C. 35°C to 42°C
D. -10°C to 110°C

12. **The reading of a clinical thermometer does not fall on its own when taken out of the mouth because:**

A. the glass is thick
B. mercury is very heavy
C. the kink stops the mercury flowing back
D. it is a digital device

13. **A patient's temperature is recorded as 40°C. Compared with the normal body temperature, this shows a fever of:**

A. 3°C above normal
B. no fever
C. 3°C below normal
D. 40°C above normal

14. **In an experiment, equal wax balls on copper, iron and glass rods fall at 20 s, 45 s and 90 s. The best conductor of heat is:**

A. all are equal
B. iron
C. glass
D. copper

Section B — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. **Assertion (A):** Metals are used to make cooking utensils.
Reason (R): Metals are good conductors of heat.

2. **Assertion (A):** On a sunny day, a car with dark-coloured seats becomes hotter inside than one with light-coloured seats.
Reason (R): Dark surfaces are good absorbers of heat radiation.

3. **Assertion (A):** Birds often fluff up their feathers in cold weather.
Reason (R): The air trapped between the feathers is a good conductor of heat.

Answer Key

Multiple Choice (Higher Order)

1. B	**2.** A	**3.** D	**4.** C	**5.** A
6. B	**7.** B	**8.** A	**9.** B	**10.** C
11. D	**12.** C	**13.** A	**14.** D	

Assertion–Reason

1. A
2. A
3. C